Helical Imperative — Architecture Series
Technical Note

Omega-Gated KV Eviction for Bounded-Memory LLM Sessions on Consumer Hardware

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Abstract

We describe a session governance method that maintains semantic coherence over arbitrarily long conversations within a fixed token budget. A structural coherence score (Omega, Ω) computed against a session axis declared at t=0 governs eviction of context entries. The axis is hash-committed at session open and never updated; all subsequent context is scored by deviation from this reference. We present an open implementation against Qwen2.5-14B running locally on Apple M3 hardware (32 GB unified memory) and an empirical comparison against a sliding-window baseline across six systematic tests over 20 conversation turns. Key results: (1) governed throughput exceeded naive baseline by 35.3% (37.5 vs 27.7 tokens/s) sustained over 20 turns due to bounded prefill cost; (2) budget enforcement held at zero violations across all turns; (3) a planted fact survived six noise turns under Omega eviction but was lost under sliding-window eviction; (4) the eviction system reached steady state at 57.3% eviction rate by turn 19 (89 units accumulated, 38 selected), with tokens fed stabilizing at ~1990 from turn 3 onward. All code and test harness are released under CC0.

1 Motivation

Recent demonstrations of 400-billion parameter Mixture-of-Experts models running on mobile hardware via SSD-streaming inference (Flash-MoE, March 2026) establish that frontier-scale models can operate on consumer devices. The binding constraint shifts from model weight capacity to context management: as sessions grow, the cost of feeding full conversation history to the model during prefill rises linearly, degrading throughput and eventually exhausting available memory.

Sliding-window eviction — discarding the oldest turns once a fixed token limit is reached — is the standard mitigation. It is structurally blind: a fact planted early in a session is evicted regardless of its relevance to the session's governing question. We propose an alternative: eviction governed by a coherence score computed against the session's founding axis, such that semantically load-bearing context is retained while topically orthogonal content is preferentially evicted.

This note describes the method, its implementation, and empirical results on a 14B-parameter model running at interactive speeds on commodity hardware.

2 Method

2.1 Axis Declaration

At session open (t = 0), the text of the first user message is hash-committed as the session axis:

```
axis_hash = sha256(first_user_message)
```

This hash is immutable. All subsequent scoring is deviation from this reference, not from the current conversational state. The axis cannot be updated mid-session; doing so would constitute drift ratification.

2.2 Context Segmentation

Conversation turns are parsed into ContextUnits by an argument AST segmenter. Segmentation boundaries are detected at logical switch markers (contrastive connectives, causal bridges, domain transitions). Each unit carries: role (user/assistant), turn index, stack depth (nesting level of qualifications), cross-switch flag (opens with a causal bridge across a prior argument boundary), and a token estimate.

Units with cross_switch = True are assigned PROVISIONAL evidence tier, capping their maximum solvency contribution. Short units (< 20 tokens) are UNAUDITED. All others are CALIBRATED.

2.3 Omega Scoring

Each unit is scored against the session axis by a heuristic coherence estimator:

```
overlap = |axis_words ∩ unit_words| / |unit_words|
structural_Ω = 1.0 - min(overlap × 3.0, 1.0)
composite = (1 - Ω) × recency_weight × role_weight × sol
```

where recency_weight decays with a half-life of 10 turns, role_weight = 1.0 for user turns and 0.85 for assistant turns, and sol is the solvency coefficient of the evidence tier (CALIBRATED = 0.90, PROVISIONAL = 0.70, UNAUDITED = 0.50). Low composite score indicates a candidate for eviction.

The current scorer is lexical (word-overlap against axis). Section 5 describes the planned upgrade to embedding-distance scoring, which is required for technically dense text where semantic proximity is not lexically surfaced.

2.4 Window Construction

At each turn, units are ranked by composite score. The latest user turn is pinned (never evicted; causality invariant). Remaining units are selected greedily in descending composite order until the token budget is exhausted. The resulting window is fed to the model; the full conversation history is retained in a local store for re-scoring on subsequent turns.

2.5 Governance Invariants

Five structural invariants are enforced:

- I1 (Causality): governance precedes the governed event; the window is constructed before the model call.
- I2 (Conservation): token budget is finite and tracked; FIN on exhaustion.
- I3 (Separation): the scorer is architecturally separate from the generator.
- I4 (Irreversibility): the session audit log is append-only.
- I5 (Identity): the axis is declared at t=0, hash-committed, and never updated.

3 Implementation

The implementation consists of three modules:

- context_stack.py — Ingestion layer. Argument AST segmenter, ContextUnit dataclass, IEP window formula.
- stp_session.py — Transport layer. Axis declaration, Omega scorer, window builder, append-only audit log.
- omega_session.py — Generation layer. Ollama /api/chat client, session loop, CLI.

All dependencies are standard library Python except the Ollama local inference server. No external ML libraries are required for the v0 lexical scorer. Total implementation is approximately 400 lines across three files.

Hardware: Apple M3, 32 GB unified memory. Model: Qwen2.5-14B, 4-bit quantized, served via Ollama. Operating system: macOS.

4 Empirical Results

A six-test harness was run against the live Ollama instance. Tests T1 and T2 are unit tests requiring no model inference. Tests T3 through T6 exercise the full inference loop.

Test	Result	Note
`T1: Segmentation`	**PASS**	6 units, 1 provisional, IEP window 600
`T2: Ω scoring + eviction`	**PASS**	on-topic Ω=0.295, off-topic Ω=0.688, pin invariant
`T3: t/s stability`	**PASS**	Governed 37.5 t/s vs baseline 27.7 t/s (+35.3%, 20 turns)
`T4: Budget enforcement`	**PASS**	0 violations over 20 turns, budget 800 tokens
`T5: Early recall`	**PASS**	Governed True, sliding window False
`T6: Eviction curve`	**PASS**	89 units → 38 selected at turn 19, 57.3% eviction rate

4.1 Throughput (T3)

Over 20 turns of synthetic conversation, the governed system achieved a mean of 37.5 tokens/s versus 27.7 tokens/s for the naive baseline (+35.3%). The improvement arises because the governed system bounds the prompt fed to the model at each turn; as conversation grows, baseline prefill cost rises linearly while governed prefill cost remains bounded at the configured budget. The tokens-fed trace confirms this: after an initial ramp (15 tokens at turn 0, 1617 at turn 2), the window stabilizes between 1975-2000 tokens for all subsequent turns, never requiring a full history pass. The effect compounds over longer sessions.

4.2 Budget Enforcement (T4)

With an 800-token budget over 20 turns, zero budget violations were recorded across all turns. The pinned-turn slack (20 tokens) was never exceeded. This confirms that the window construction algorithm correctly enforces the budget as a hard ceiling regardless of session length.

4.3 Early Recall (T5)

A specific technical fact — the energy threshold for Szilard's engine, kT ln2 joules per bit — was planted in the first two turns of a conversation, followed by six topically unrelated turns (population of Tokyo, inventor of the telephone, definition of photosynthesis). A recall probe was then issued.

Under Omega-gated eviction, the planted fact was retained and correctly recalled. Under sliding-window eviction (last 4 turns only), the fact was lost and not recalled. This is the primary empirical claim of this note: axis-coherent eviction preserves semantically load-bearing early context that sliding-window eviction discards.

4.4 Eviction Curve (T6)

Over 20 turns with a 1500-token budget, the session accumulated 89 total units while the selected window stabilized between 26-38 units. Final eviction rate at turn 19 was 57.3% (51 of 89 units evicted). The eviction curve reveals a characteristic steady-state behavior: eviction rate climbs steeply through turns 7-11 as the budget fills, then levels off as the rate of new unit arrival roughly matches the rate of low-Ω unit displacement. Tokens fed stabilized at ~1490-1500 from turn 7 onward, confirming that the budget ceiling is maintained across an extended session. The budget was never exceeded in any turn.

Turn	Total	Selected	Evicted	Tokens Fed
0	1	1	0	15
1	7	7	0	617
2	11	10	1	1,481
3	12	11	1	1,495
4	17	15	2	1,453
5	19	16	3	1,461
6	24	17	7	1,486
7	34	23	11	1,500
8	39	25	14	1,483
9	41	26	15	1,494
10	45	26	19	1,481
11	51	26	**25**	1,495
12	53	31	**22**	1,476
13	60	29	**31**	1,494
14	64	29	**35**	1,483
15	67	31	**36**	1,500
16	71	33	**38**	1,488
17	79	34	**45**	1,494

Turn	Total	Selected	Evicted	Tokens Fed
18	80	35	**45**	1,481
19	89	38	**51**	1,499

Table 2: Eviction curve over 20 turns. Bold evicted values indicate turns where eviction rate exceeded 40%.

5 Limitations and Future Work

5.1 Lexical Scorer

The v0 Omega scorer uses word-overlap against the axis text. This fails for technically dense content where semantic proximity is not lexically expressed — for example, a conversation about NANOGrav gravitational wave detection where the axis is established using domain vocabulary that may not recur verbatim in later turns. The correct upgrade is embedding-distance scoring: replace word-overlap with cosine distance between sentence embeddings (e.g., all-MiniLM-L6-v2, 22 MB). Embedding distance correctly captures domain continuity regardless of surface vocabulary.

The threshold structure for embedding-based switching is:

```
distance < 0.20    same argument, elaboration        no switch
0.20 - 0.35        related argument, same domain     soft switch
0.35 - 0.55        domain transition                 hard switch
> 0.55             orthogonal content                evict candidate
```

5.2 Single-Session Scope

The current implementation governs a single session with a single axis. Multi-session memory (axis recovery from prior conversations), hierarchical axis structures (sub-sessions inheriting a parent axis), and cross-session coherence tracking are not implemented.

5.3 Token Estimation

Token count is estimated as character_length / 4. A tokenizer call would give exact counts at the cost of a dependency on the model-specific tokenizer. For budget enforcement purposes the approximation is conservative.

5.4 Recall Measurement

The T5 recall test uses keyword presence (kT ln2) as the recall signal. A more rigorous evaluation would use semantic similarity between the planted fact and the model's response. The keyword test is sufficient for a proof-of-concept but should be replaced with embedding similarity for publication-quality evaluation.

6 Prior Art and Open Release

The governance architecture described here — axis declaration at t=0, ratio-based viability scoring, elastic context transition costs, re-entrant audit — was first articulated in the Helical Imperative framework beginning December 2025, with progressive deposits to Zenodo under ORCID 0000-0001-8054-2085. The STP (Solvency Transport Protocol) and IEP (Internal Ear Protocol) formalisms underlying this implementation are documented in prior deposits at that ORCID.

This note and all accompanying code are released under CC0 1.0 Universal. No rights reserved. The implementation is intended as a reproducible reference for the method, not a proprietary claim. Researchers are invited to fork, extend, and cite.

References

[1] Anemll (@anemll). Running 400B model on iPhone. Twitter/X, March 23, 2026.

[2] Woods, D. Flash-MoE: SSD-streaming inference for Mixture-of-Experts models. GitHub, 2026. github.com/Anemll/flash-moe

[3] Apple Research. LLM in a Flash: Efficient Large Language Model Inference with Limited Memory. arXiv:2312.11514, 2023.

[4] Mohan, D.T. Helical Imperative Architecture Series, Notes 1-24. Zenodo, 2025-2026. ORCID 0000-0001-8054-2085.

[5] Qwen Team. Qwen2.5 Technical Report. arXiv:2412.15115, 2024.